Applied Deep Learning

Introduction

Starting at 6:30

About Me – Robert Ross

aka, Rob.

Not, Sir, Professor, or any other honorific



https://www.linkedin.com/in/robertjohnross/

Senior Lecturer – School of Computer Science
Lead – Cognitive Systems Research Group
Director – Applied Intelligence Research Centre

Active National Funding
- RI Funded Adapt Research Centre
- Enterprise Ireland Funded Innovation Partnerships and Com Funds
- Co-Leader, Precision-ALS Project for Motor Neuron Disease

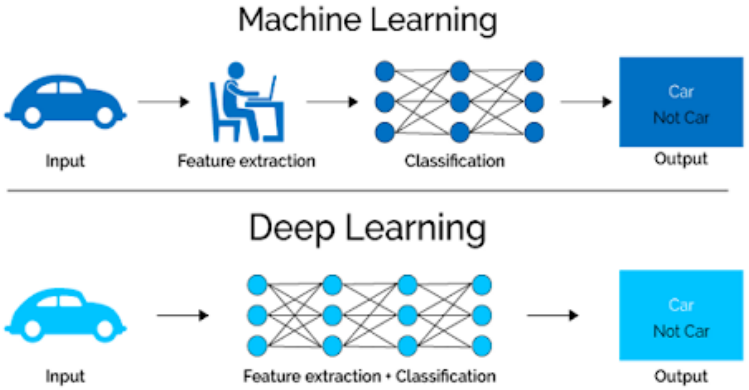
Research Areas:
- Situated Interactive Systems (Talking Robots / Games Characters)
- Applied Computer Vision Applications

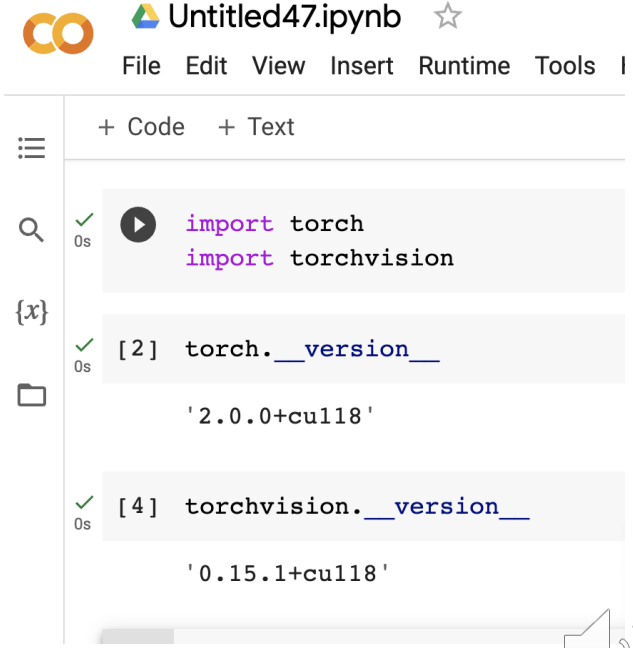
- Jumping In
- Course Scope
- Prerequisites
- Course Structure
- Assignments and Assessment
- Coding
- Lab: Model Training with Collab and VS Code



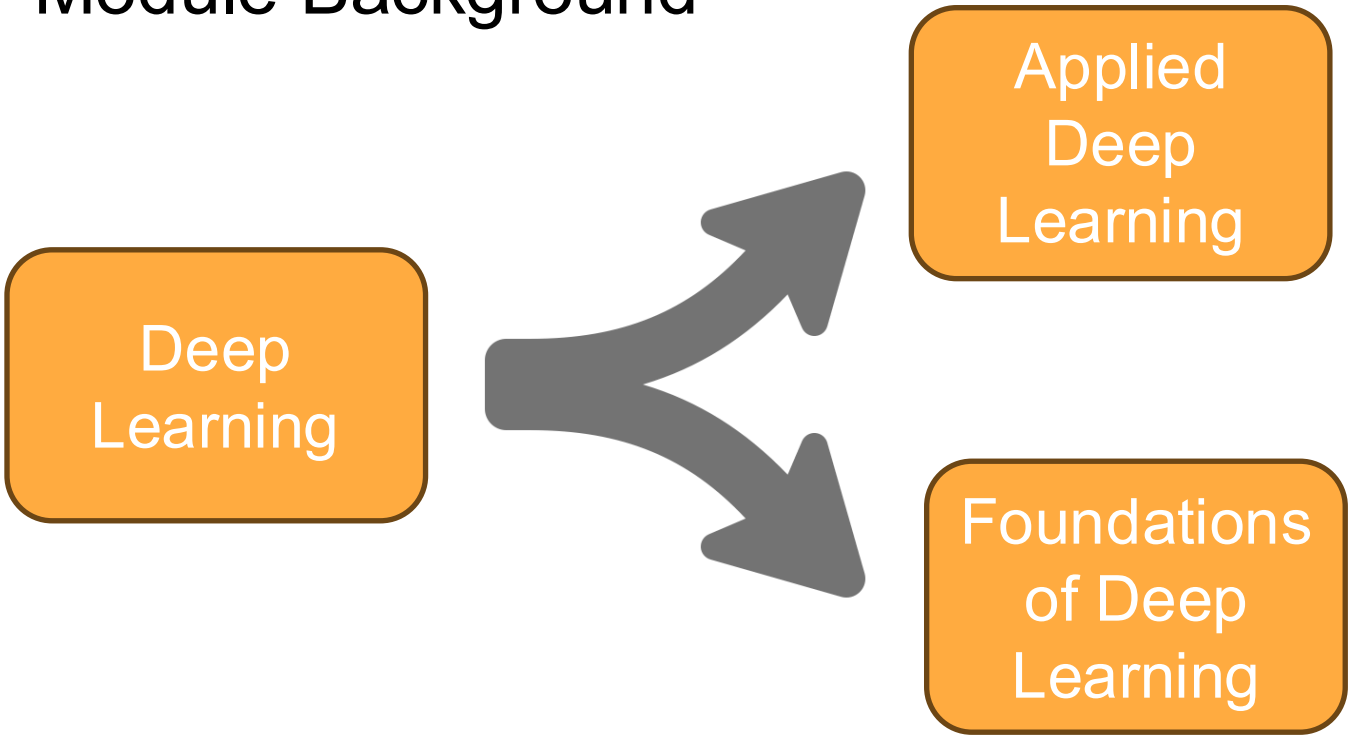
Who is this module for?

- Anyone from either a Data Analytics or Advanced Software Development background who wants to develop a practical understanding of Deep Learning – covering the training and use of the most frequently used model types





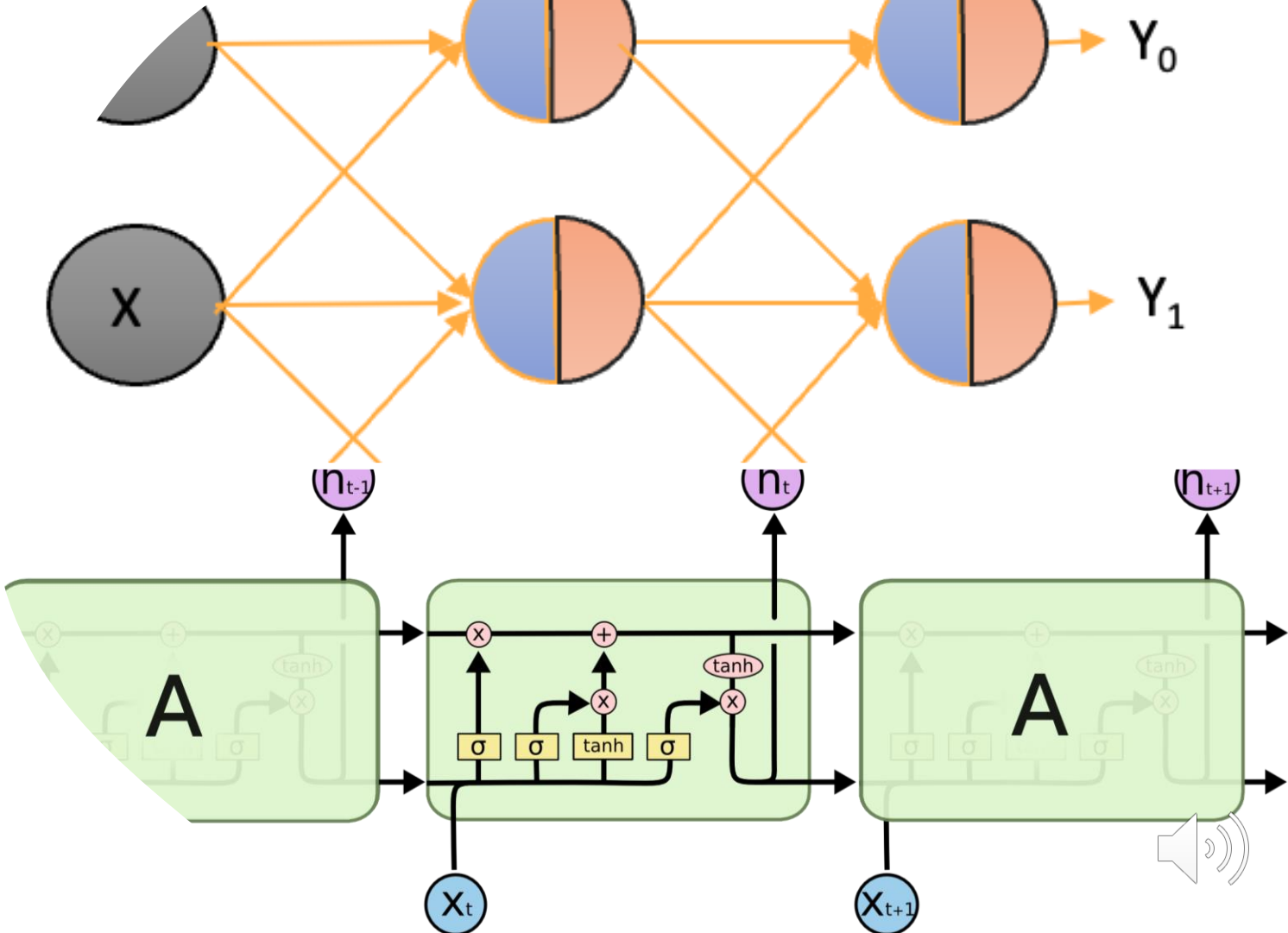
Module Background



Learning Outcomes

- *Understand the common elements and architectural constructions used in Deep Learning*

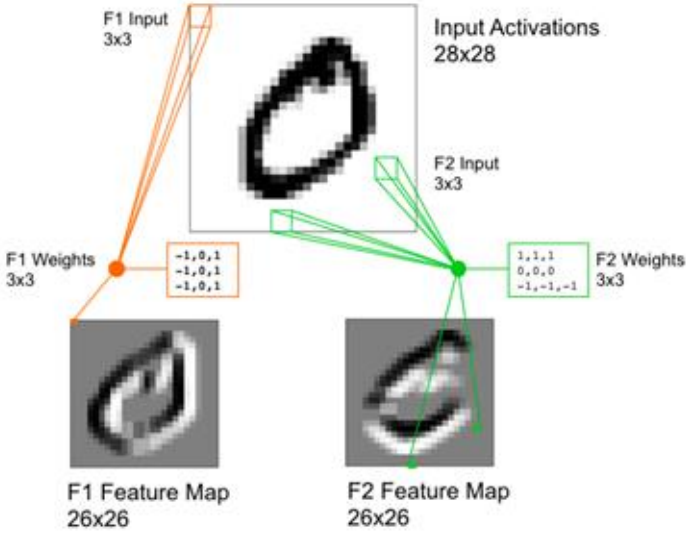



X
Y_0
Y_1
h_{t-1}
h_t
h_{t+1}
A
A
tanh
σ
σ
tanh
σ
tanh
X_t
X_{t+1}

Learning Outcomes

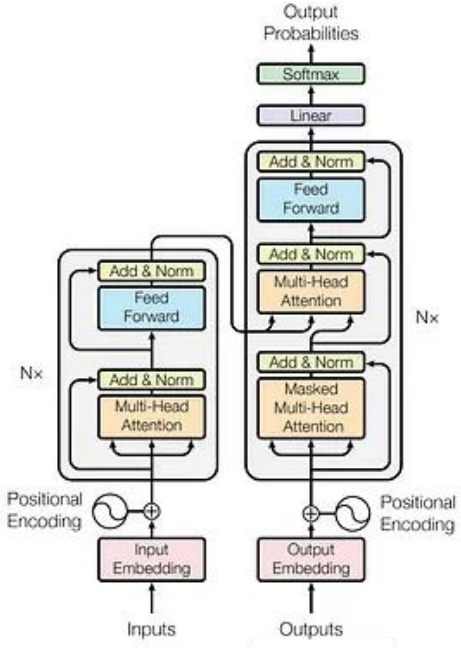
- *Understand the common elements and architectural constructions used in Deep Learning*
- *Apply and critique Deep Learning Architectures to typical deployments in areas such as image processing and text processing.*





BERT

Encoder



GPT

Decoder

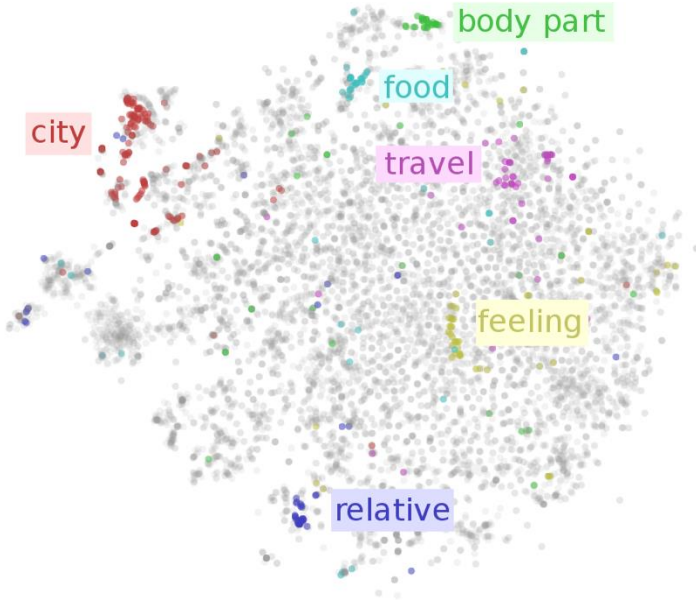
Learning Outcomes

- *Understand the common elements and architectural constructions used in Deep Learning*
- *Apply and critique Deep Learning Architectures to typical deployments in areas such as image processing and text processing.*
- *Examine the challenge of representations and data types as used in the context of Deep Learning.*



One-hot encoding

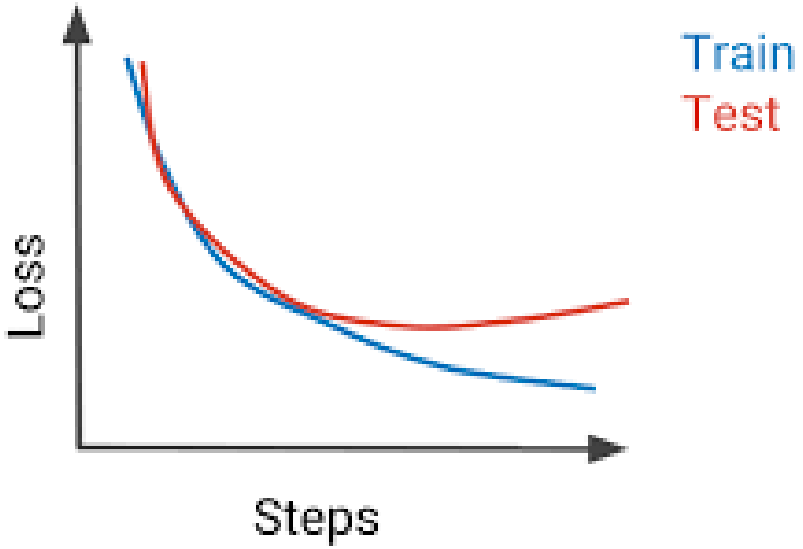
	cat	mat	on	sat	the
the =>	0	0	0	0	1
cat =>	1	0	0	0	0
sat =>	0	0	0	1	0
...	...				



Learning Outcomes

- *Understand the common elements and architectural constructions used in Deep Learning*
- *Apply and critique Deep Learning Architectures to typical deployments in areas such as image processing and text processing.*
- *Examine the challenge of representations and data types as used in the context of Deep Learning.*
- *Examine the challenge of overfitting and data bias in the context of Deep Learning solution development.*



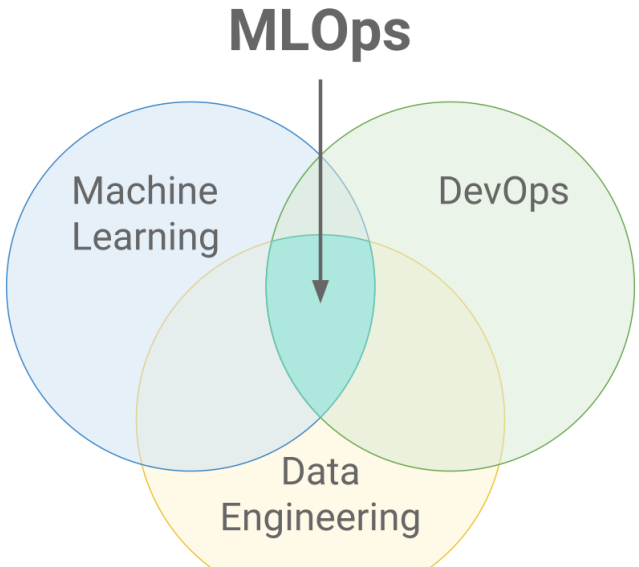

Train
Test
Loss
Steps

Learning Outcomes

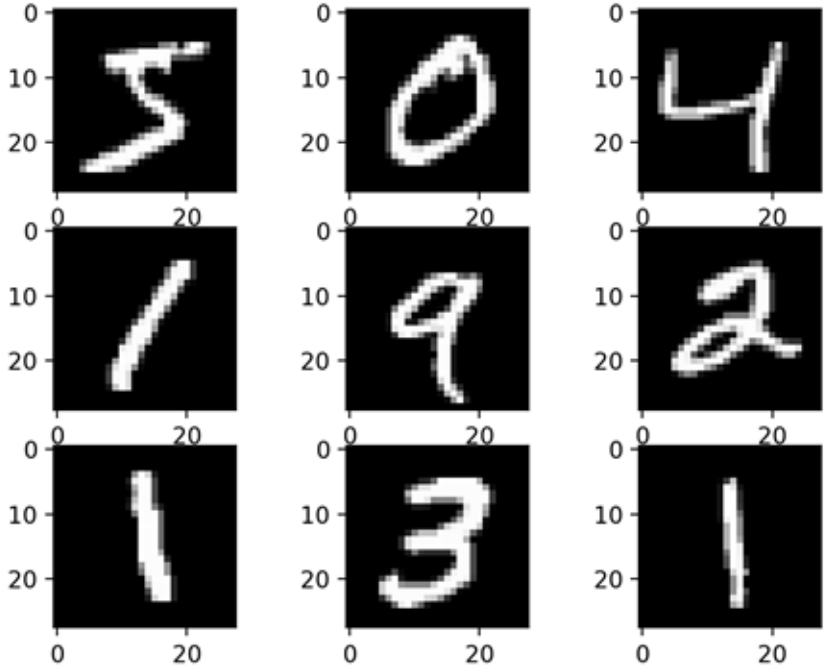
- *Understand the common elements and architectural constructions used in Deep Learning*
- *Apply and critique Deep Learning Architectures to typical deployments in areas such as image processing and text processing.*
- *Examine the challenge of representations and data types as used in the context of Deep Learning.*
- *Examine the challenge of overfitting and data bias in the context of Deep Learning solution development.*
- *Describe and critique common techniques for deploying and training deep neural networks.*




ROG
GAME ON


MLOps
Machine Learning
DevOps
Data Engineering

Jumping In: Mnist Data



Jumping In: Accurate Digit Classification in 14 Lines

```
import tensorflow as tf

mnist = tf.keras.datasets.mnist
(x_train, y_train),(x_test, y_test) = mnist.load_data()

x_train = tf.keras.utils.normalize(x_train, axis=1)
x_test = tf.keras.utils.normalize(x_test, axis=1)

model = tf.keras.models.Sequential()
model.add(tf.keras.layers.Flatten())
model.add(tf.keras.layers.Dense(128, activation=tf.nn.relu))
model.add(tf.keras.layers.Dense(128, activation=tf.nn.relu))
model.add(tf.keras.layers.Dense(10, activation=tf.nn.softmax))

model.compile(optimizer="adam", loss="sparse_categorical_crossentropy", metrics=["accuracy"])
model.fit(x=x_train, y=y_train, epochs=5)

test_loss, test_acc = model.evaluate(x=x_test, y=y_test,verbose=0)
print('\nTest accuracy: ' + str(test_acc*100) + "%" )
```

Jumping In: Training

```
Epoch 1/5
1875/1875 [==============================] - 5s 2ms/step - loss: 0.4656 - accuracy: 0.8687
Epoch 2/5
1875/1875 [==============================] - 3s 2ms/step - loss: 0.1113 - accuracy: 0.9657
Epoch 3/5
1875/1875 [==============================] - 3s 2ms/step - loss: 0.0738 - accuracy: 0.9763
Epoch 4/5
1875/1875 [==============================] - 3s 2ms/step - loss: 0.0491 - accuracy: 0.9857
Epoch 5/5
1875/1875 [==============================] - 3s 2ms/step - loss: 0.0375 - accuracy: 0.9884

Test accuracy: 97.28000164031982%
```

```python
import torch
from torchvision import datasets, transforms
from torch import nn, optim

train_loader = torch.utils.data.DataLoader(
    datasets.MNIST('./data', train=True, download=True,
                   transform=transforms.ToTensor()),
    batch_size=64, shuffle=True)
test_loader = torch.utils.data.DataLoader(
    datasets.MNIST('./data', train=False, download=True,
                   transform=transforms.ToTensor()),
    batch_size=1000)

model = nn.Sequential(
    nn.Flatten(),
    nn.Linear(28*28, 128),
    nn.ReLU(),
    nn.Linear(128, 10)
)
optimizer = optim.Adam(model.parameters(), lr=0.001)
loss_fn = nn.CrossEntropyLoss()

for _ in range(5):
    for x, y in train_loader:
        optimizer.zero_grad()
        loss = loss_fn(model(x), y)
        loss.backward()
        optimizer.step()

correct = sum((model(x).argmax(1) == y).sum().item() for x, y in test_loader)
total = len(test_loader.dataset)
print(f'Test Accuracy: {100 * correct / total:.2f}%')

```

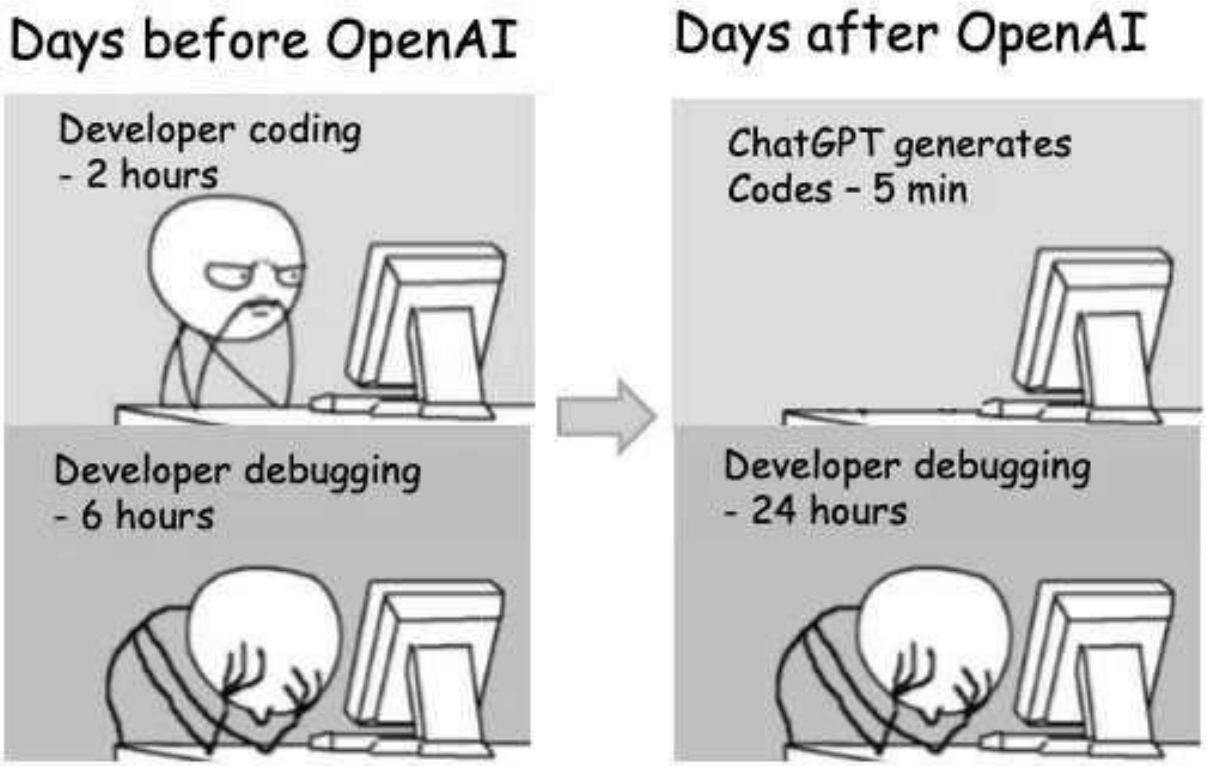
Course Scope



Course Scope

- Goal: Study key concepts in Deep Learning
  - Emphasis on understanding and genuine skill building
  - Making sure you know what is in front of you is correct and high quality
    - (anyone can build a bad presentation)




Days before OpenAI
Developer coding
- 2 hours
Developer debugging
- 6 hours
Days after OpenAI
ChatGPT generates
Codes - 5 min
Developer debugging
- 24 hours

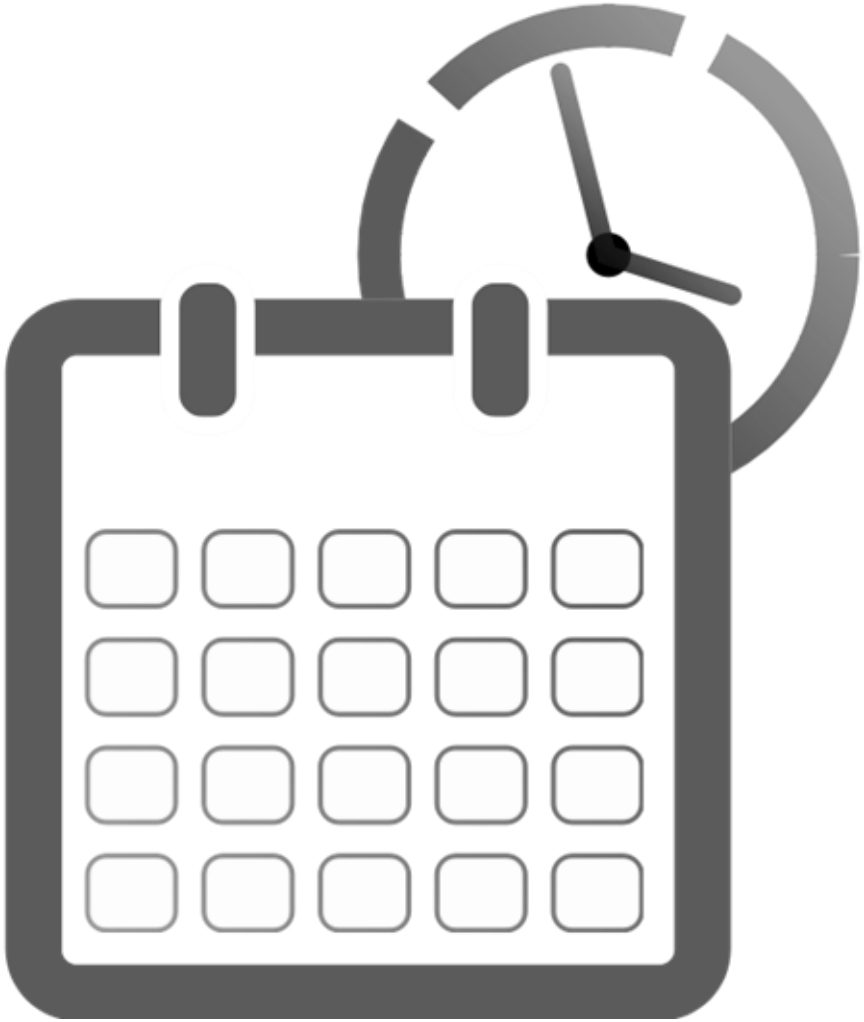
Course Scope

- Goal: Study key concepts in Deep Learning
  - Emphasis on understanding and genuine skill building
- Lecture Notes:
  - Slides with some Jupyter Notebooks (not 100% consistent!)
- Not Covered
  - The history of neural networks
  - The biological inspiration of neural networks
  - Explainability, Ethics and other Human-Centric issues
- Learning Resources
  - Course is mostly self contained
  - Links to external resources will be given
- At MSc and PhD Level - Self directed study is assumed!

Prerequisites

- Students **should ideally** have completed undergraduate or Masters level modules on:
  - Artificial Intelligence or
  - Machine Learning
- Taking Machine Learning as a co-requisite is acceptable but continued study is important
- Consult online resources
- Ask Gen AI for Assistance – but force yourself to learn!!!!
- Assignments to be coded in Python
- Laptop
  - I strongly encourage all students to set up their own python environments to learn
  - Lab Today on VS Code
- Willingness to interact
  - We don’t all like groups, trust me! But they are a good way to learn.

Course Structure



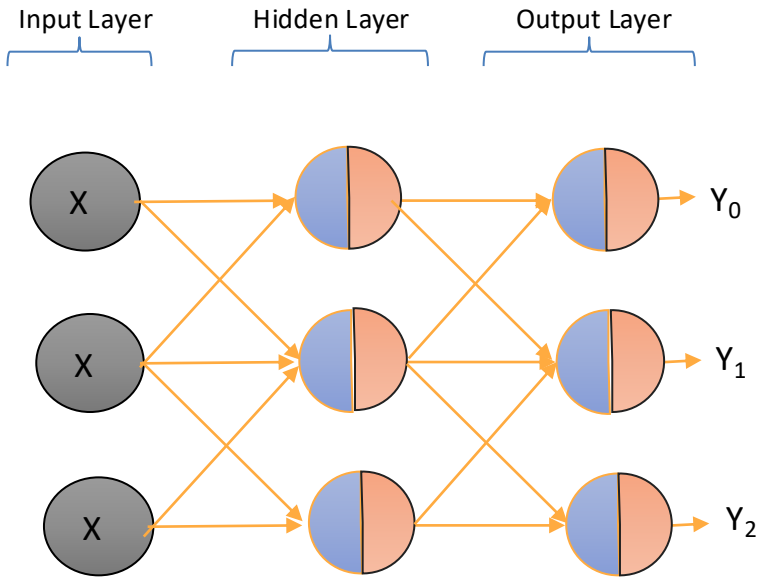
Week by Week (always subject to change)

- Week 1 – Introduction
- Week 2 – Neural Network Essentials
- Week 3 – Training and Loss Functions
- Week 4 – Better Activations
- Week 5 – Preventing Overfitting
- Week 6 – Convolutional Neural Networks
- Week 7 – Review Week (Bank Holiday)
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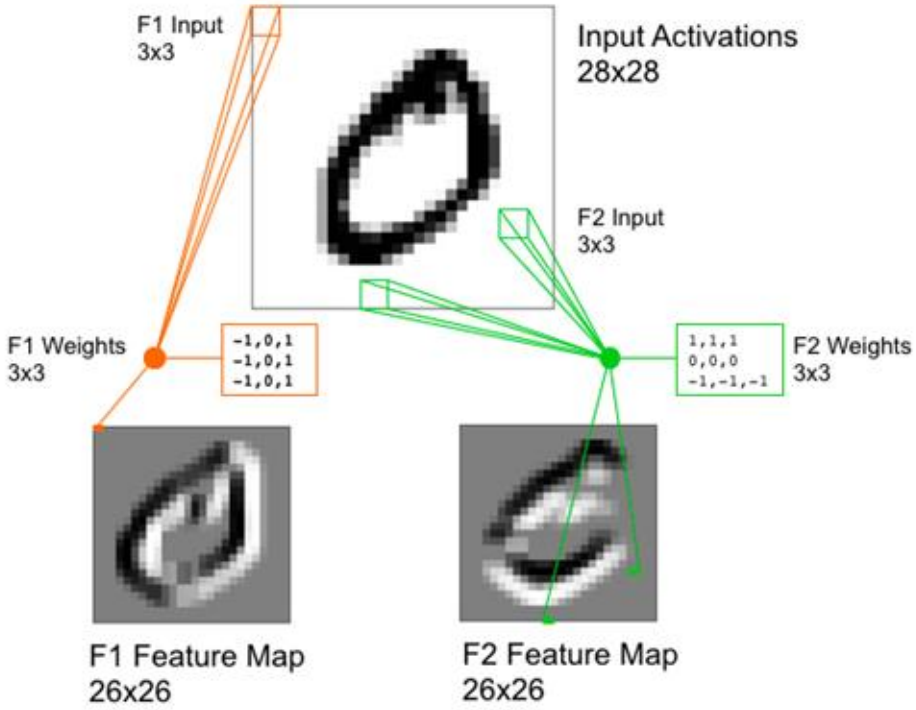
Basic Neural Networks



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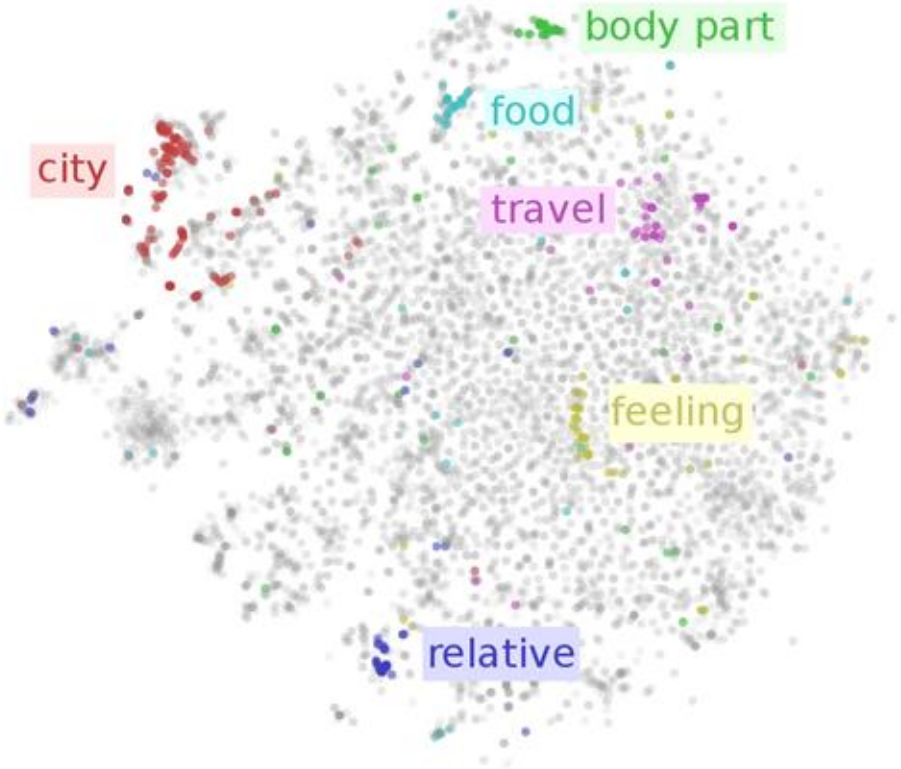
Edge Detectors, Filters and Features



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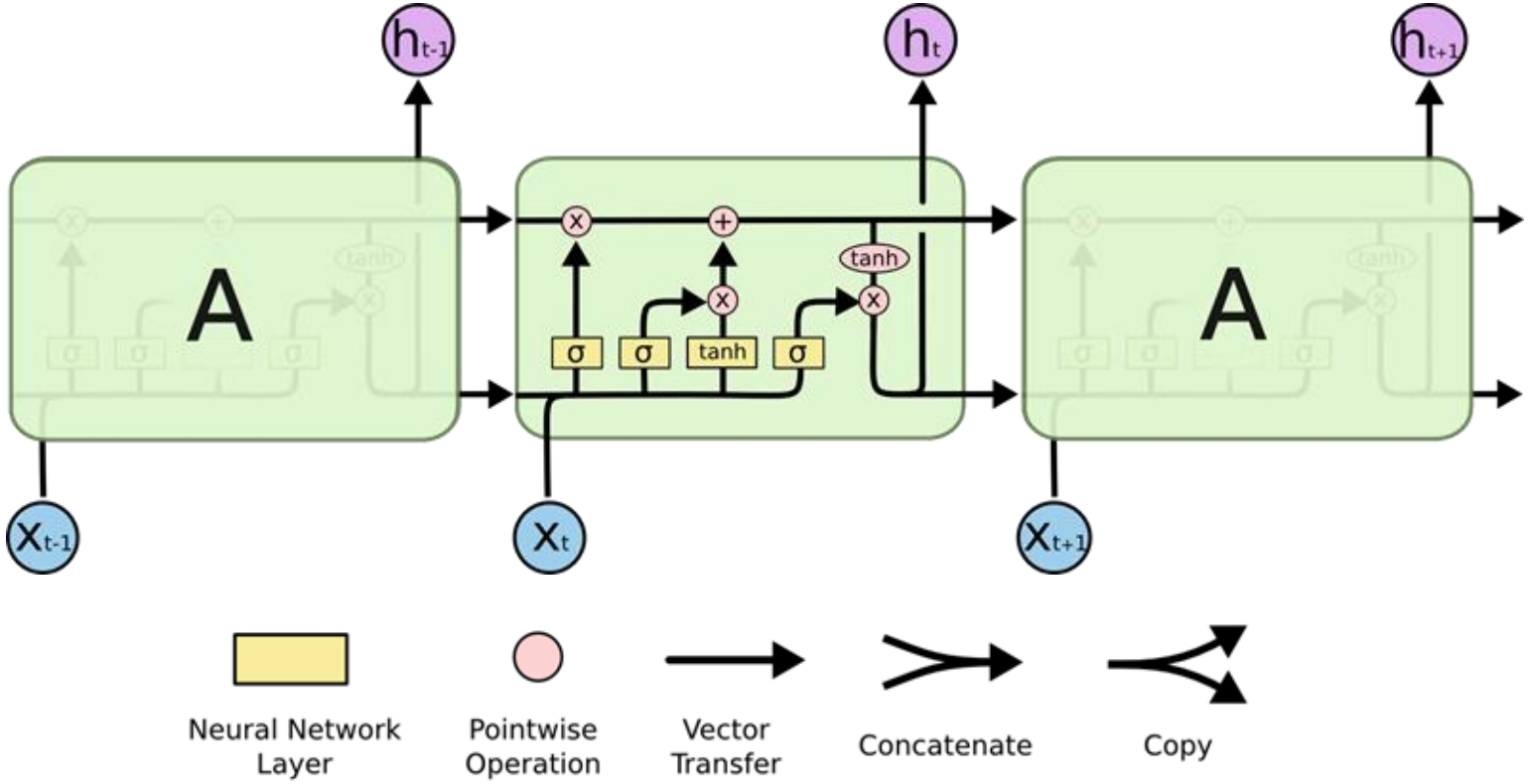
Meaning Spaces



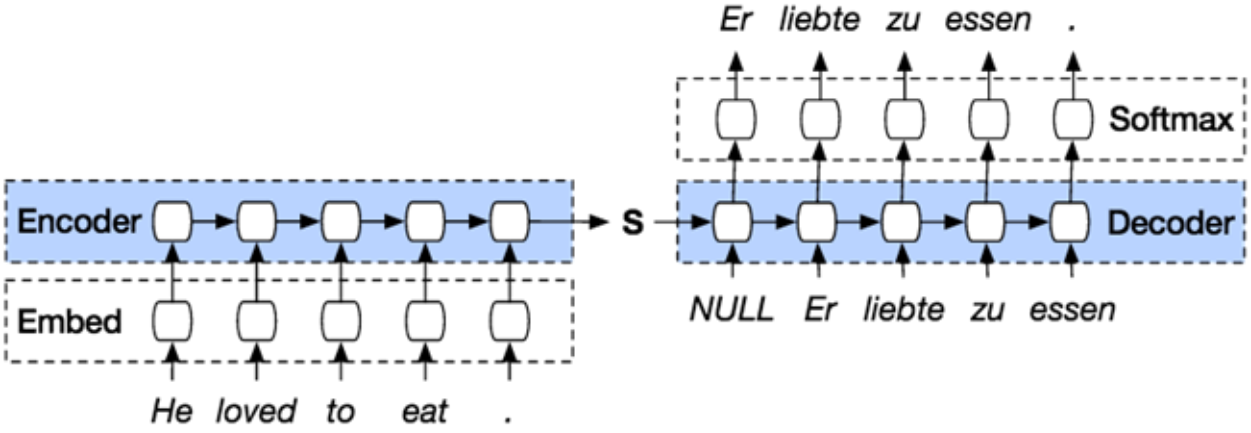
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LSTM Based RNN



LSTM Applications - Translation



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Transformers



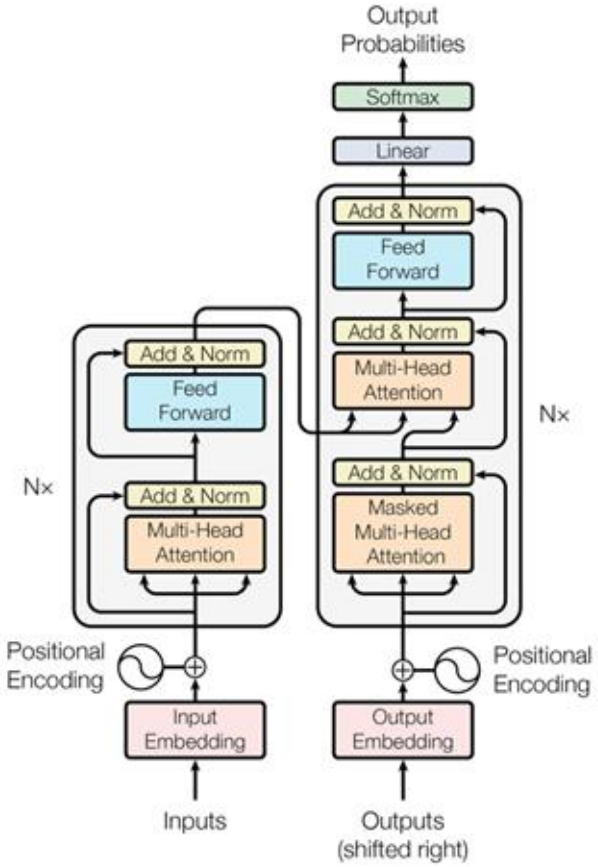


Figure 1: The Transformer - model architecture.

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Hardware Optimization

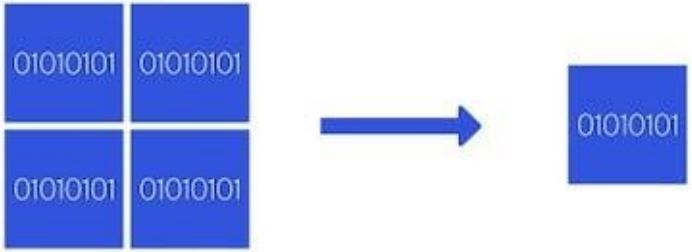


Quantization

Floating point → **Integer**

3452.3194 → 3452

32 bit → **8 bit**



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Delivery

- 5 ECTS
- Three hours contact per week
- A sometimes Flipped Classroom Model
  - If you get notes in advance -- read them in advance!!!!
- Assessment
  - End of Semester Exam: 50%
  - Individual Project: 50%

Coding

- Python plus requested libraries and frameworks
  - Use Virtual Environments -- pip, I suggest avoiding Anaconda
  - Discussed in the Lab Session Today
- Examples
  - Frameworks used
  - Small amounts of code
  - Sometimes Tensorflow / Sometimes PyTorch
- Server Environment
  - Most code can be run locally
  - Google Collab can generally be used for most experimental work
  - VS Code + Github Copilot is probably what most will use if you have a reasonable machine

Content Presentation

Lab for Weeks 1 and 2

Part 1:
Download the notebook for week 1.
Upload it onto google Colab and execute content
Switch Runtime model on Google Colab between GPU and CPU. Can you see any difference? Probably not for such a simple model.

Part 2:
Install Python 3.12 on your own machine
Make sure Python is being added to your PATH
Run python from the command line and verify the version
Install VS Code / VS Codium and set up a project

